

Therefore, the area of the board = $(40)(12) = 480 \text{ cm}^2$

6

D

After the stone leaves the hand of the astronaut, it does not experience any horizontal forces anymore. The only force acting on the stone is its own weight acting vertically downwards to the surface of the Moon. This force is constant as there is no opposing air resistive force acting on the stone due to no atmosphere present. Hence, option D.

7

B

$$P_{\text{friction}} = F_{\text{friction}} v$$

